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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD2001-1 – FUNDAMENTALS OF MACHINE LEARNING
(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which type of machine learning involves labeled data?
A) Supervised learning B) Unsupervised learning
C) Reinforcement learning D) Intelligence Learning
2. What is the main objective of the theory of generalization in machine learning?
A) To minimize training time B) To ensure the model performs well on unseen data
C) To reduce model complexity D) To optimize hardware usage
3. What is the cost function used in linear regression?
A) Cross-entropy loss B) Mean squared error
C) Hinge loss D) Log loss
4. What does reinforcement learning primarily rely on?
A) Labeling data B) Maximizing rewards
C) Minimizing variance D) Reducing noise
5. Which activation function is used in logistic regression?
A) Sigmoid B) ReLU
C) Tanh D) Softmax
6. What is the purpose of the margin in an SVM?
A) To define the decision boundary B) To minimize the cost function
C) To maximize the distance between support vectors D) To reduce overfitting
7. Which of the following is an assumption of the Naive Bayes classifier?
A) Features are dependent on each other B) The model requires a large amount of data
C) Features are dependent on each other D) Features are independent of each other given the class
8. What is the primary goal of deep learning?
A) To design shallow networks B) To improve clustering accuracy
C) To reduce the number of hidden layers D) To process large amounts of data with complex models
9. What is the main purpose of an artificial neural network (ANN)?
A) To mimic the human brain for pattern recognition B) To perform logical reasoning
C) To store large datasets D) To classify text documents only

10. What is the main purpose of an artificial neural network (ANN)?
 A) Input layer B) Output layer
 C) Bias D) Weights
11. What is the main idea behind the K-nearest neighbor (KNN) algorithm?
 A) It builds a model using a decision tree B) It predicts based on the majority class of nearest neighbors
 C) It reduces data dimensions D) It uses linear regression to classify data
12. What is the key feature of locally weighted regression?
 A) It uses the entire dataset for predictions B) It uses a fixed global model
 C) It assigns higher weights to nearby points D) It assumes data is linearly separable
13. What is the purpose of cross-validation in machine learning?
 A) To reduce the number of features B) To evaluate the model's performance on unseen data
 C) To increase the training data size D) To optimize hyperparameters
14. Which evaluation measure calculates the average error between predicted and actual values?
 A) Precision B) Recall
 C) Mean Squared Error (MSE) D) Accuracy
15. What does a p-value in hypothesis testing represent?
 A) The probability of accepting the null hypothesis when it is false B) The likelihood that the null hypothesis is true
 C) The likelihood of observing the data given the null hypothesis is true D) The probability of rejecting the null hypothesis when it is true
16. In Q-learning, what does the Q-value represent?
 A) The quality of an action in a given state B) The distance between clusters
 C) The error rate in predictions D) The probability of a hypothesis being correct
17. Which of the following is an output of hierarchical clustering?
 A) A confusion matrix B) A dendrogram
 C) A scatter plot D) A regression line
18. Which distance metric is commonly used in K-means clustering?
 A) Manhattan distance B) Euclidean distance
 C) Cosine similarity D) Jaccard similarity
19. What is overfitting in decision trees caused by?
 A) Too few features B) Too many splits or deep trees
 C) High learning rates D) Lack of labeled data
20. What is the splitting criterion in a decision tree?
 A) Cost function B) Learning rate
 C) Information gain or Gini index D) Gradient descent

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Explain the following with suitable examples:
- Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning

Marks BT* CO* PO*

8 L*2 1 1

- b) Write the Find-S algorithm and apply it to the given dataset (Fig. Q1(b)) for the target attribute **"Play Tennis."**

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Mild	High	Weak	Yes
4	Overcast	Mild	Normal	Weak	Yes
5	Overcast	Cool	Normal	Weak	Yes

Fig. Q1(b)

2. a) Explain the different applications of machine learning.
 b) Design an algorithm for candidate elimination concept with suitable example
3. a) With suitable example explain the following
 i. Theory of Generalization.
 ii. Learning Curve.
 b) Explain multivariate regression with the suitable example.

8	L3	2	1
8	L2	1	1
8	L3	2	1
8	L2	1	1
8	L2	2	1

Unit – II

4. a) Explain the need for logistic regression in classification problems and how the sigmoid function helps in binary classification. Illustrate with relevant data.
 b) What are Gain and Entropy in a Decision Tree, and how are they used for classification? Calculate the entropy of the given training examples with respect to the target function classification. Apply the information gain formula to evaluate attribute **"a2"** based on these training examples.

8	L2	3	1
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Instance	a1	a2	classification
1	T	T	+
2	T	T	+
3	T	F	-
4	F	F	+
5	F	T	-
6	F	T	-

Fig. Q4(b)

5. a) Explain the working of Hidden Markov Models (HMMs) with its key component.
 b) Design an algorithm for the K-Nearest Neighbors (KNN) classification technique and explain its working with suitable data sample.
6. a) Explain ANN with the suitable example
 b) Explain the Naïve Bayes Classifier and apply it to classify whether a person will Play Tennis or not for the following conditions:
 (Outlook = Overcast, Temperature = Mild, Humidity = Normal, Wind = Weak).

8	L3	3	1
8	L2	3	1
8	L3	3	1
8	L2	3	1

Use the given dataset in Fig. Q1(b)

8	L3	3	1
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Unit – III

7. a) Explain Cross validation with the suitable example.
 b) Design an algorithm for K-means and illustrate with relevant data.
8. a) With suitable example explain Q learning.
 b) Design an algorithm for Hierarchical clustering and explain with the suitable data sample.

8	L2	4	1
8	L3	5	1
8	L2	4	1
8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
